



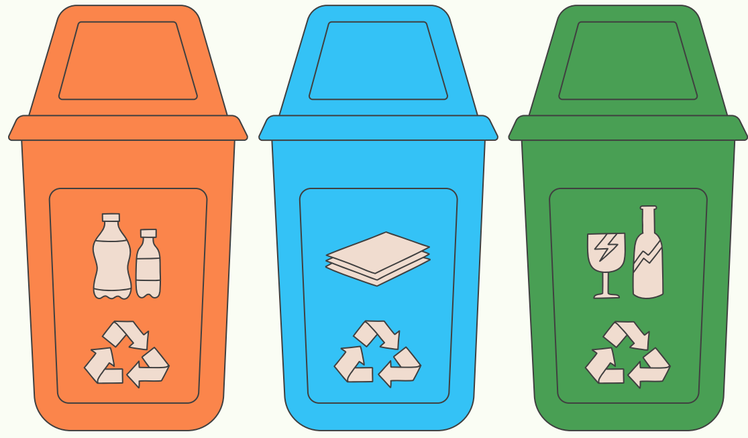


RELOOP

Smart Recycling Marketplace

Distributed web Application Project





OUR TEAM

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- Farah Akermi
- Sarah Ben Aziza
- Abderrahmen Gammoudi
- Mahdi Ben Meriem
- Amen Allah Ameri



INTRODUCTION

ReLoop: Smart Recycling Marketplace

- Connecting citizens, collectors, and recycling companies.
- Making recycling easier and more rewarding.
- Promoting the circular economy.
- Turning waste into valuable resources.



PROBLEM & PROPOSED SOLUTION

THE PROBLEM.

- Lack of connection between waste producers and recyclers
- Recyclable materials often thrown away
- Difficulties finding collection services

OUR SOLUTION.

- Centralized digital recycling marketplace
- Easy material listing and collection
- Secure transactions and environmental rewards

ACTORS

01. **Citizen:** can publish recyclable materials, browse listings, request collections, and earn points

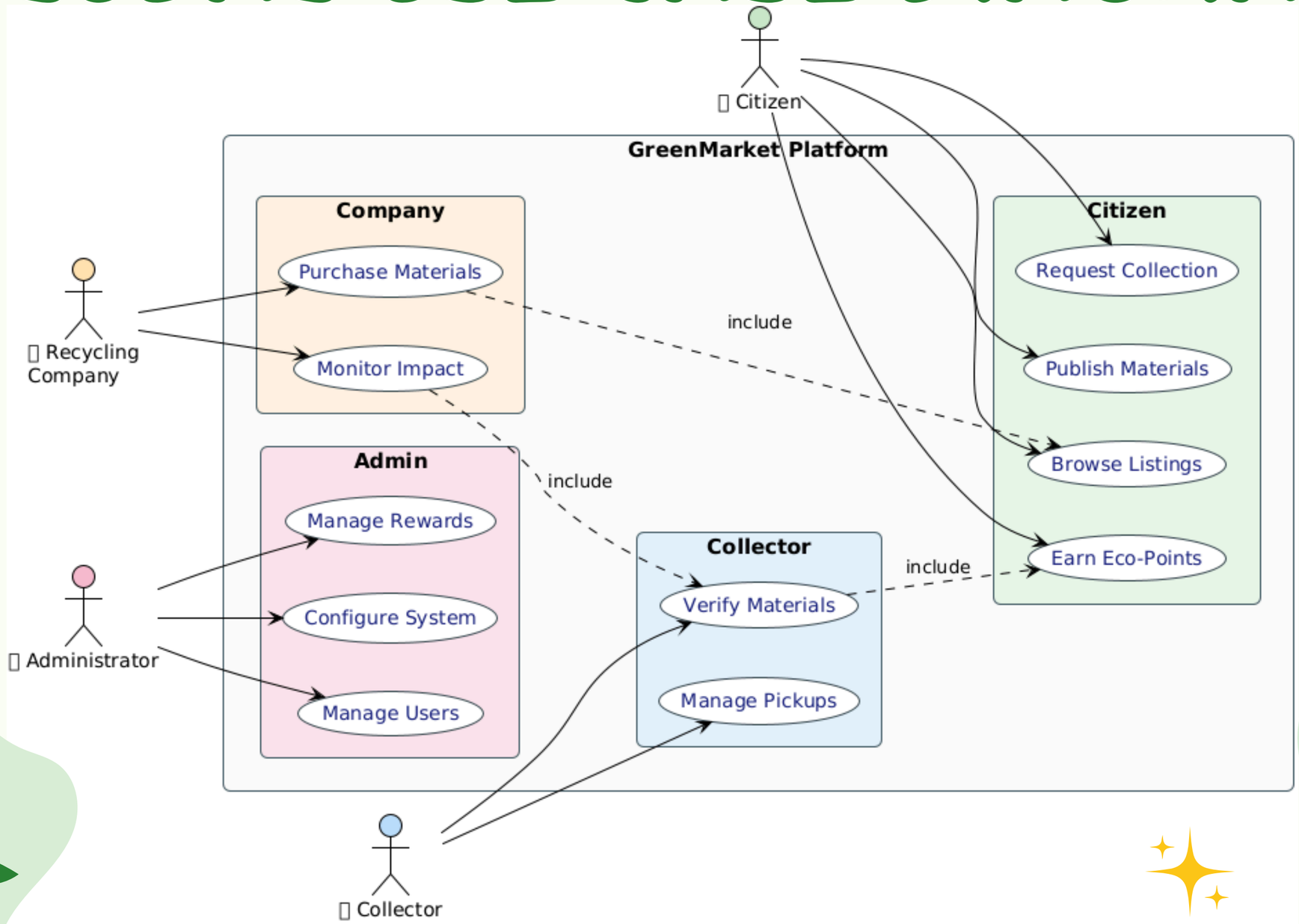
02. **Collectors**: can manage pickup requests and verify collected materials.

03. **recycling company:** can purchase recyclable materials and monitor their environmental contributions.

04. **Administrator:** can manage the platform, users, rewards, and system configurations



GLOBAL USE CASE DIAGRAM



MAIN FEATURES



- A recycling marketplace for publishing and discovering recyclable materials.
- A collection management system for scheduling and tracking pickups.
- An order and payment system for managing transactions and invoices.
- An Eco Rewards system that motivates users through points, challenges, and achievements.

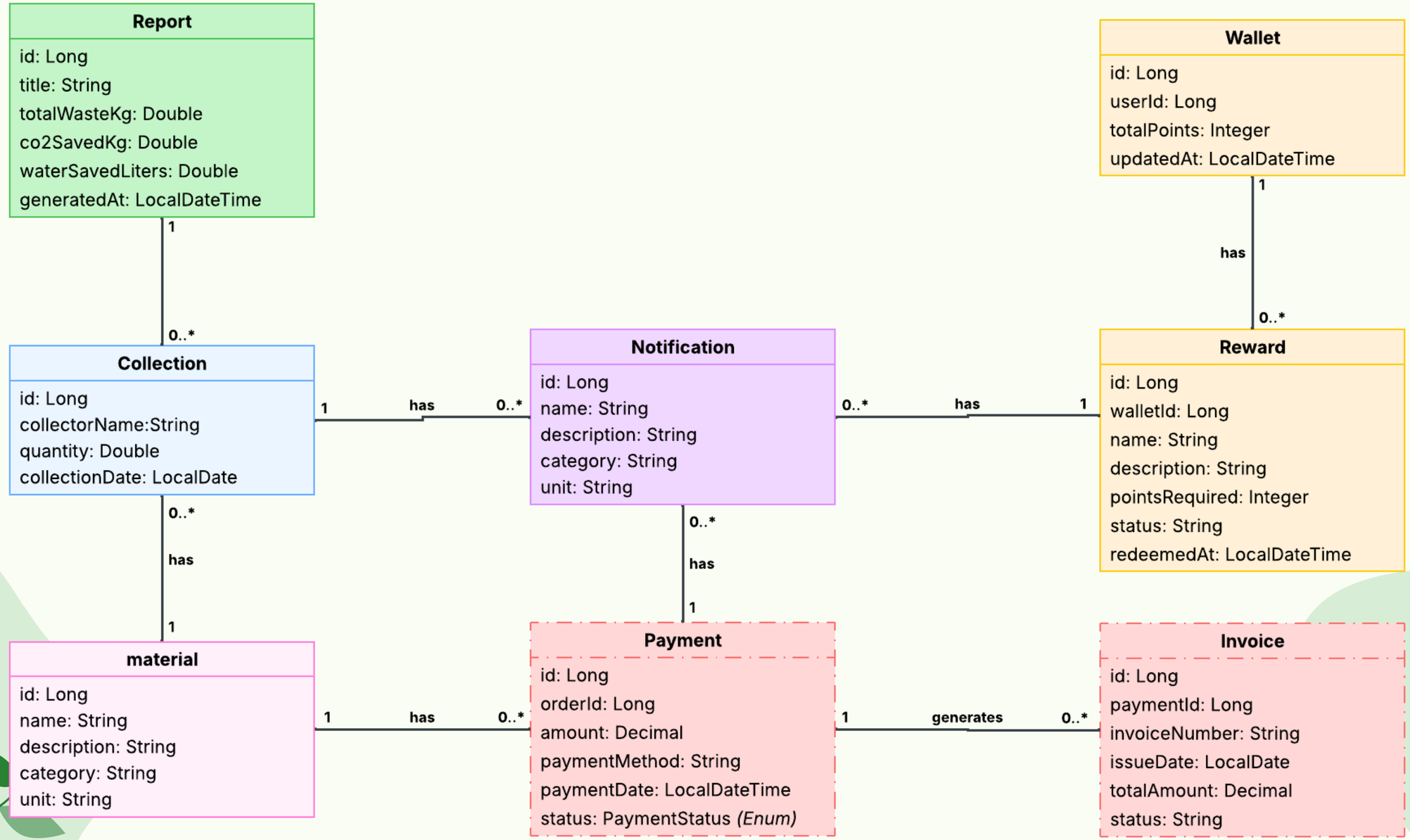
MAIN FEATURES



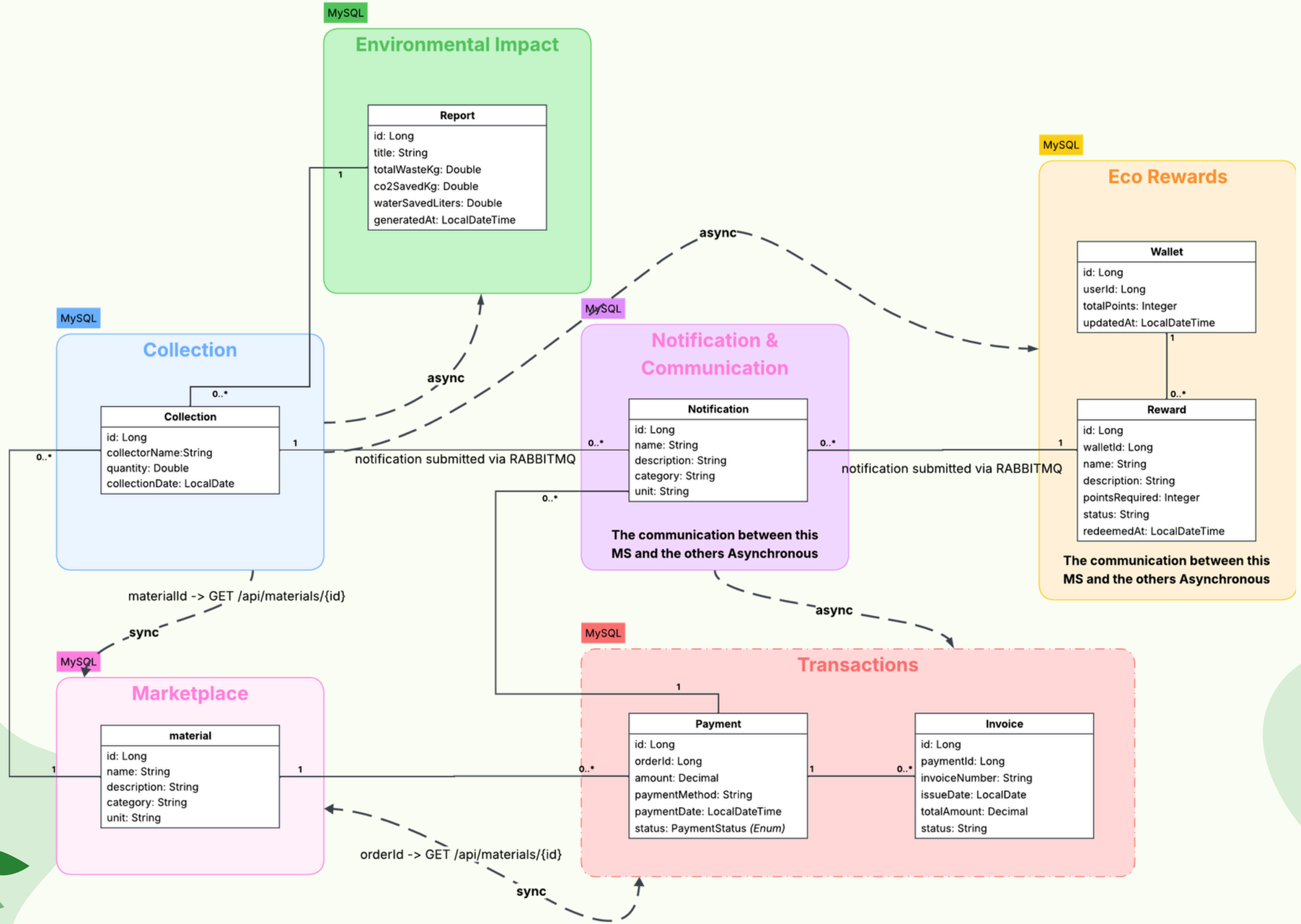
- An environmental impact system that calculates CO2 savings and waste reduction.
- A notification system that keeps users informed about important activities.
- Our collaborative AI module provides waste recognition and valuation assistance.



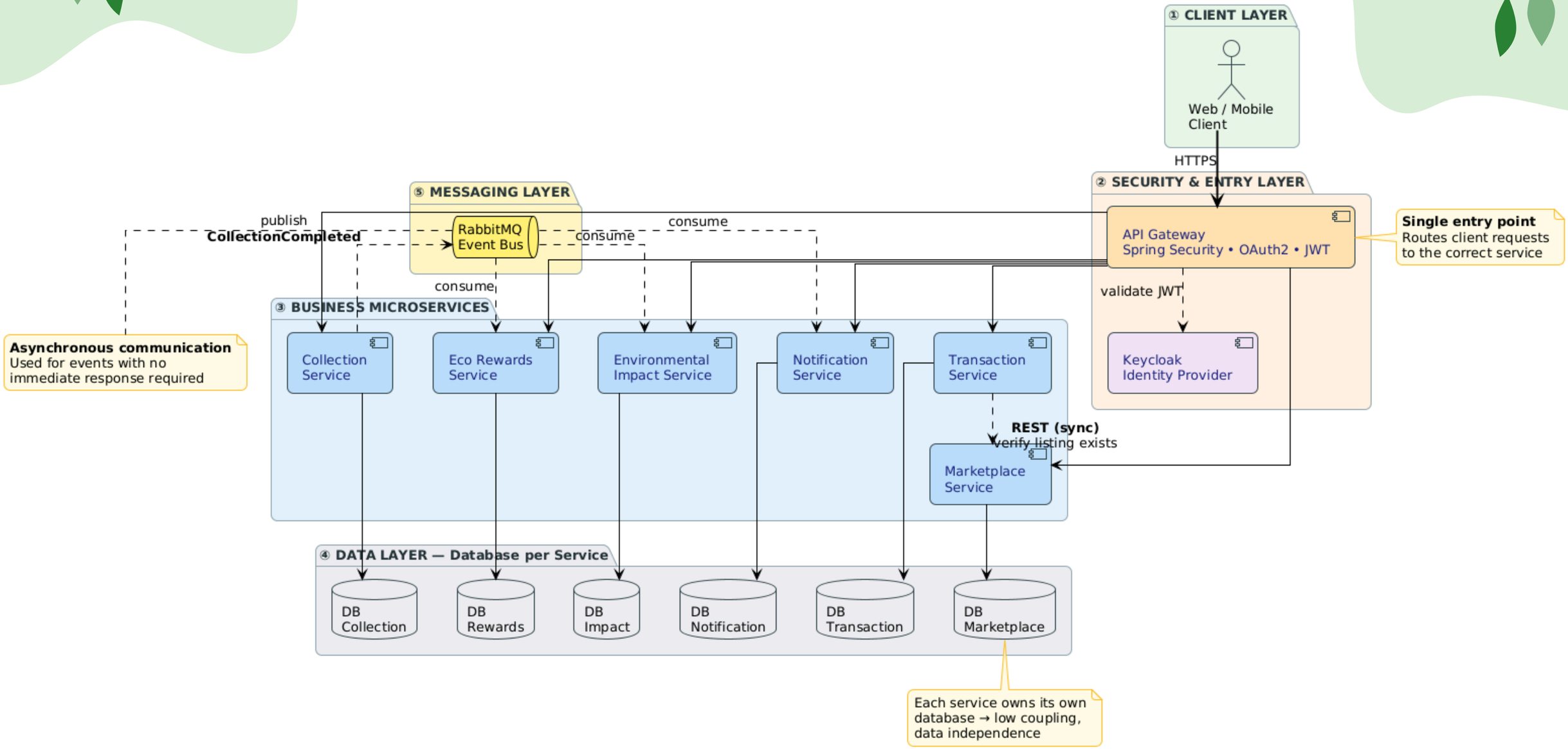
GLOBAL CLASS DIAGRAM



DOMAIN CLASS DIAGRAM



GLOBAL ARCHITECTURE



TECHNOLOGY STACK

Backend

Spring Boot

Microservices development framework

Databases

PostgreSQL

MySQL

Communication

RabbitMQ

Asynchronous messaging (events)

OpenFeign

Synchronous service-to-service calls

Security

KEYCLOAK

Spring Security

OAuth2

JWT

Identity and access management, authentication and authorization

Development Tools

Git

GitHub

IntelliJ IDEA

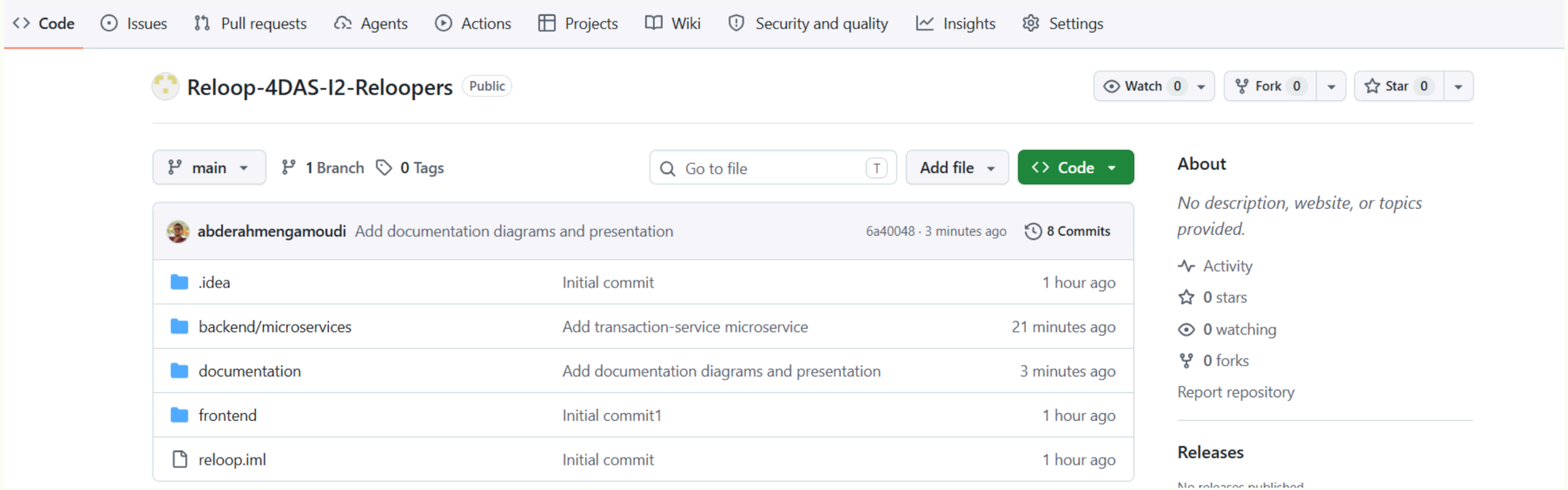
Version control, source code hosting and development environment



MICROSERVICES AND DATABASE DISTRIBUTION

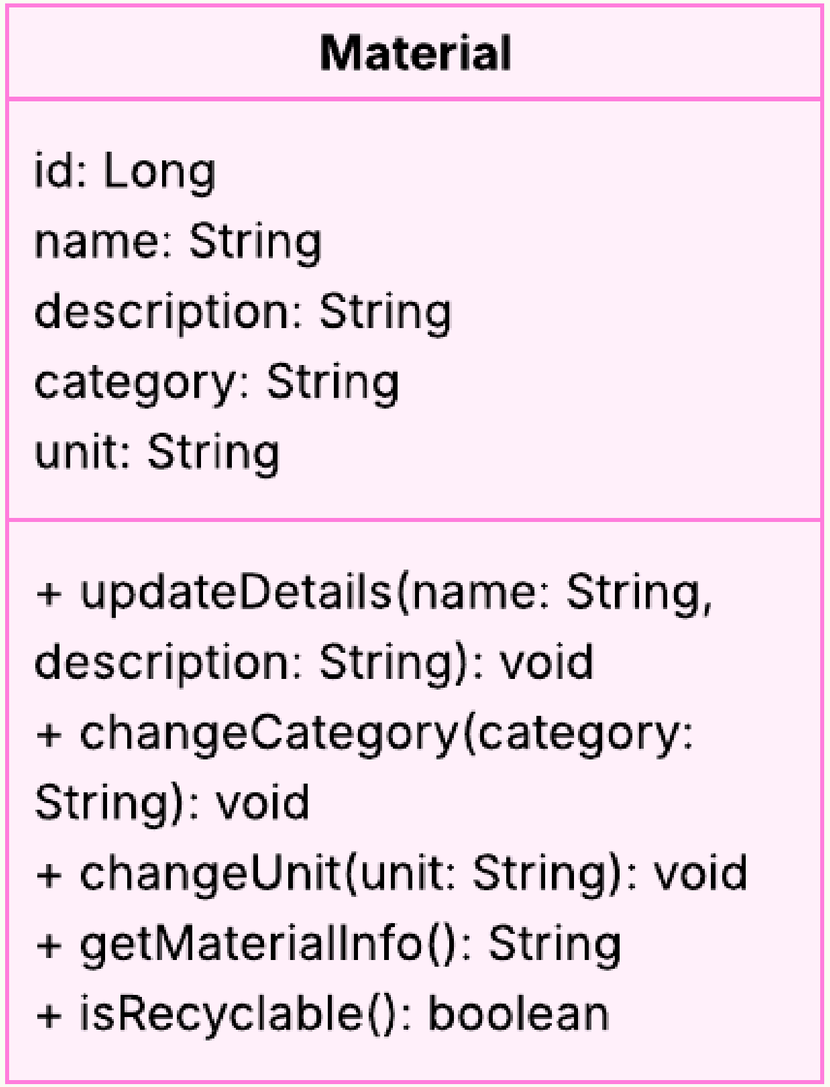
Student	Microservice	Database
1	Marketplace	MySQL
2	Collection	PostgreSQL
3	Transaction	MySQL
4	Eco Rewards	PostgreSQL
5	Environmental Impact	PostgreSQL
6	Notifications	PostgreSQL

GiTHUB & PROJECT MANAGEMENT



MICROSERVICE CLASS DIAGRAM

MARKETPLACE SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

MARKETPLACE SERVICE

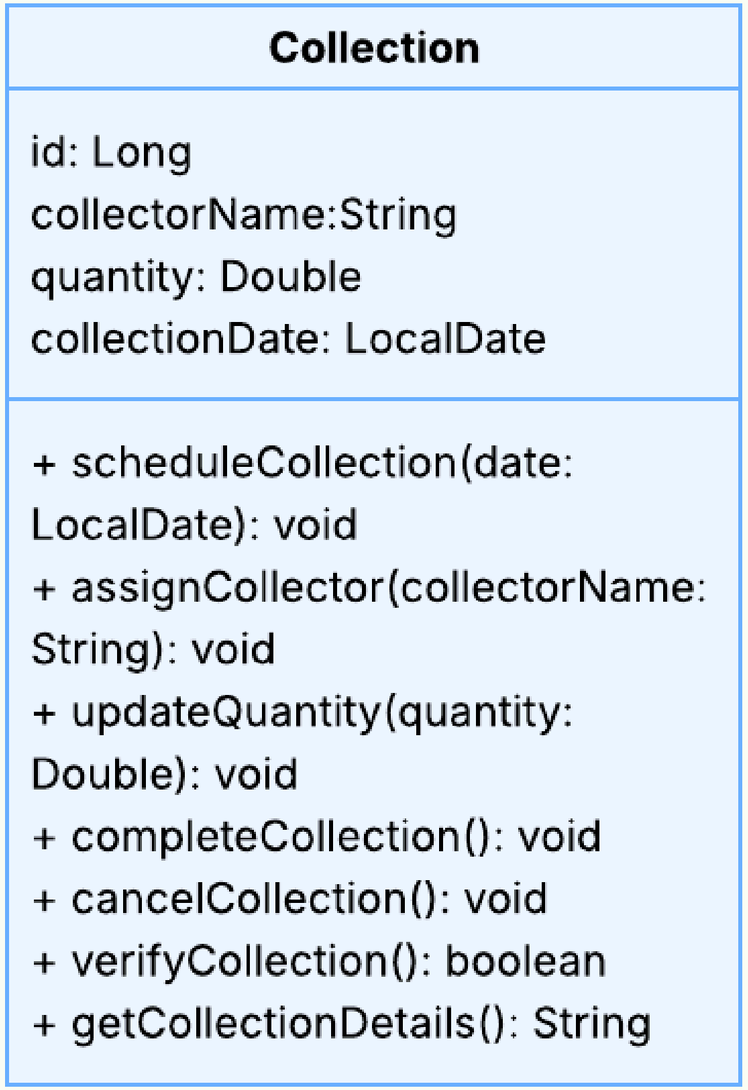
Method	Endpoint	Description	Response codes
GET	/api/materials	Retrieve all materials	200, 500
GET	/api/materials/{id}	Retrieve material details	200, 404
POST	/api/materials	Create a material	201, 400
PUT	/api/materials/{id}	Update a material	200, 400, 404

MARKETPLACE → TRANSACTION : SYNCHRONOUS (REST)

CHECK LISTING DETAILS AND AVAILABILITY BEFORE CREATING AN ORDE

MICROSERVICE CLASS DIAGRAM

COLLECTION SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

COLLECTION SERVICE

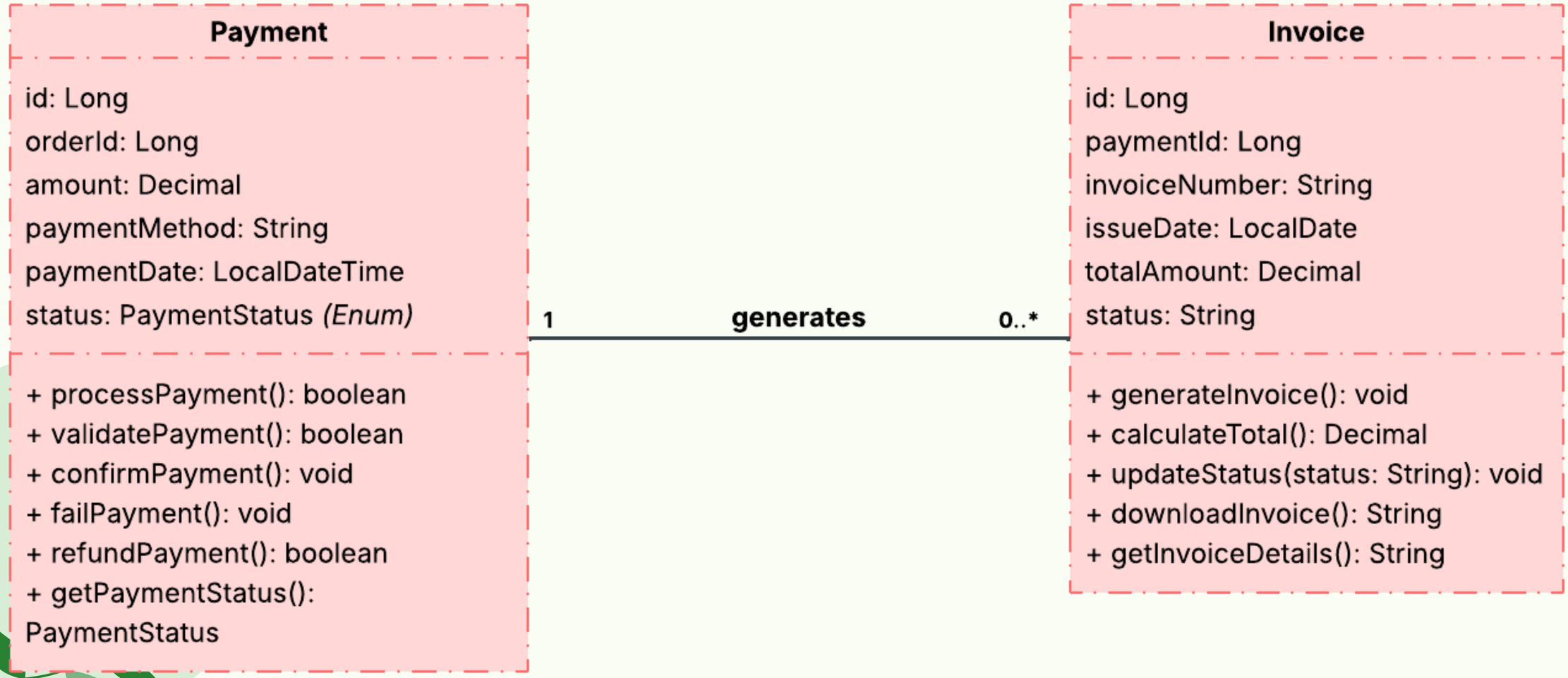
Method	Endpoint	Description	Response codes
GET	/api/collections	Retrieve all collections	200, 500
GET	/api/collections/{id}	Retrieve collection details	200, 404
POST	/api/collections	Create a collection request	201, 400
PUT	/api/collections/{id}	Update collection information	200, 400, 404

COLLECTION → MARKETPLACE : SYNCHRONOUS (REST)

RETRIEVE LISTING INFORMATION

MICROSERVICE CLASS DIAGRAM

TRANSACTIONS SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

TRANSACTION SERVICE

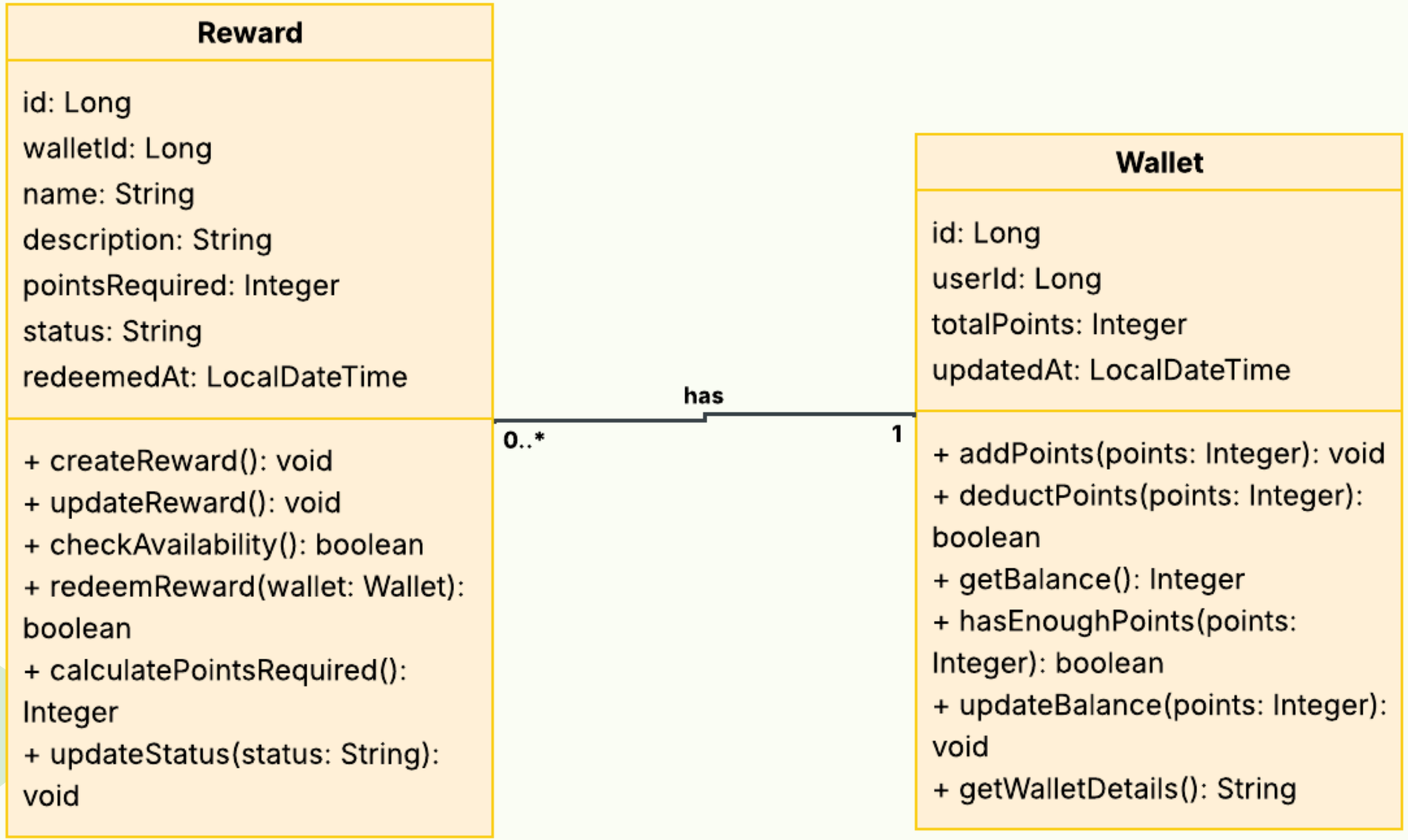
Method	Endpoint	Description	Response codes
POST	/api/payments	Process a payment	201, 400, 409
GET	/api/payments/{id}	Retrieve payment details	200, 404
GET	/api/invoices/{id}	Retrieve an invoice	200, 404

TRANSACTION → NOTIFICATION : ASYNCHRONOUS (RABBITMQ)

PUBLISH PAYMENTCOMPLETED

MICROSERVICE CLASS DIAGRAM

ECOREWARD SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

ECO REWARDS SERVICE

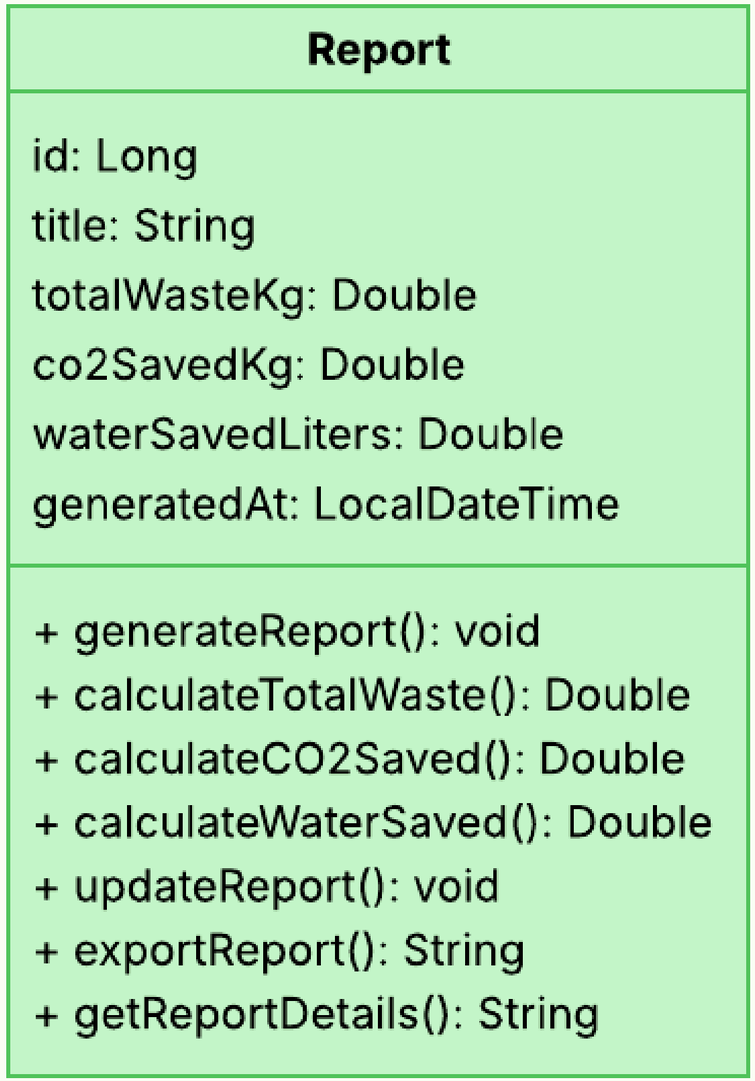
Method	Endpoint	Description	Response codes
GET	/api/rewards/wallet/{userId}	Retrieve user wallet	200, 404
GET	/api/rewards	Retrieve available rewards	200, 500
POST	/api/rewards/redeem	Redeem a reward	200, 400, 409
GET	/api/rewards/history/{userId}	Retrieve points history	200, 404

ECO REWARDS → NOTIFICATION : ASYNCHRONOUS (RABBITMQ)

PUBLISH REWARDREDEEMED

MICROSERVICE CLASS DIAGRAM

ENVIRONMENTAL IMPACT SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

ENVIRONMENTAL IMPACT SERVICE

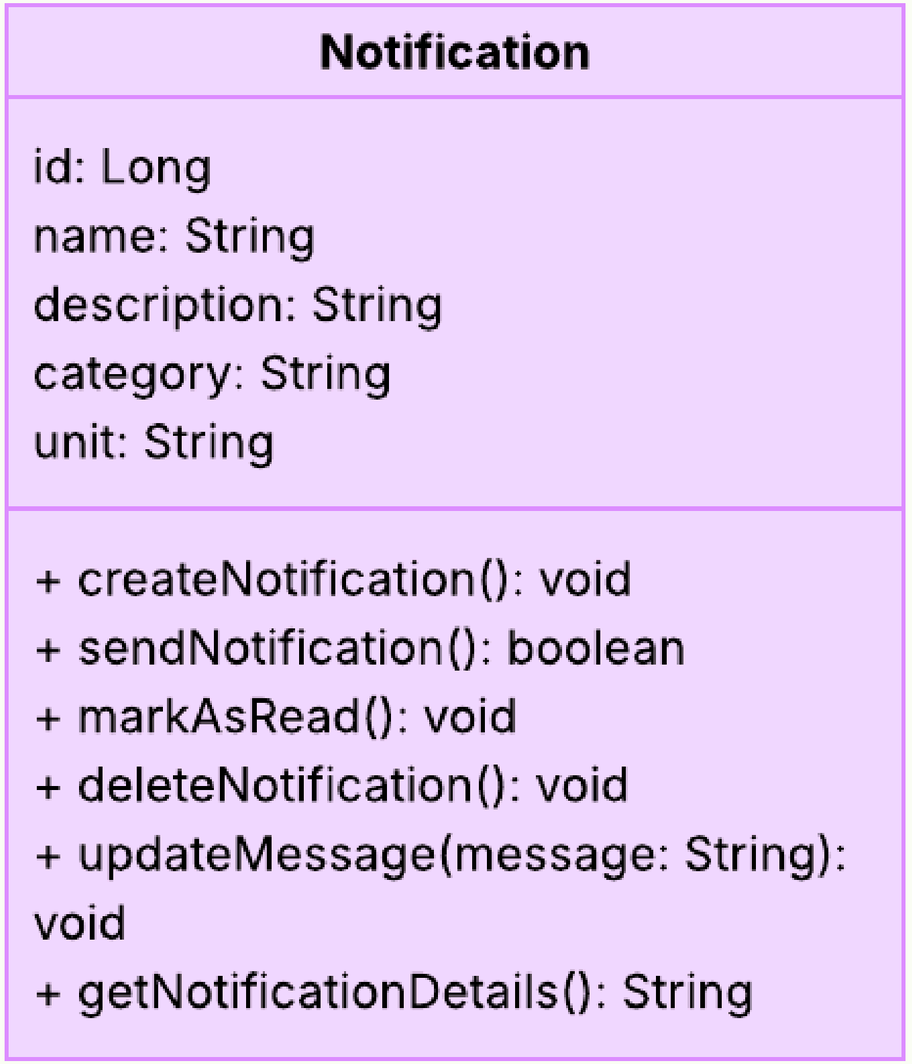
Method	Endpoint	Description	Response codes
GET	/api/reports	Retrieve sustainability reports	200,500
GET	/api/reports/{id}	Retrieve record details	200, 404
POST	/api/reports/generate	Generate a sustainability report	201,400

ENVIRONMENTAL IMPACT → NOTIFICATION : ASYNCHRONOUS (RABBITMQ)

PUBLISH IMPACTREPORTGENERATED

MICROSERVICE CLASS DIAGRAM

NOTIFICATION & COMMUNICATION SERVICE



API CONTRACTS & INTER-SERVICE COMMUNICATION

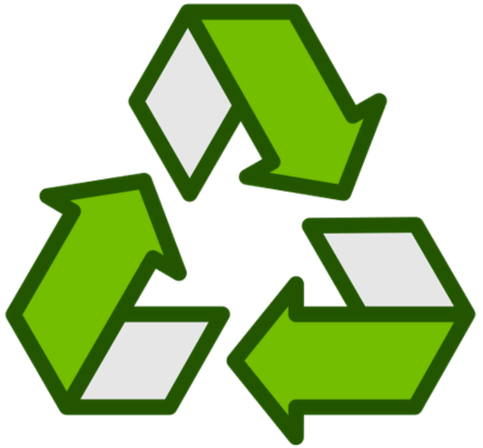
NOTIFICATION & COMMUNICATION SERVICE

Method	Endpoint	Description	Response codes
POST	/api/notifications/send	Send a notification	202, 400
GET	/api/notifications/{id}	Retrieve notification details	200, 404
GET	/api/notifications/user/{userId}	Retrieve user notifications	200, 404

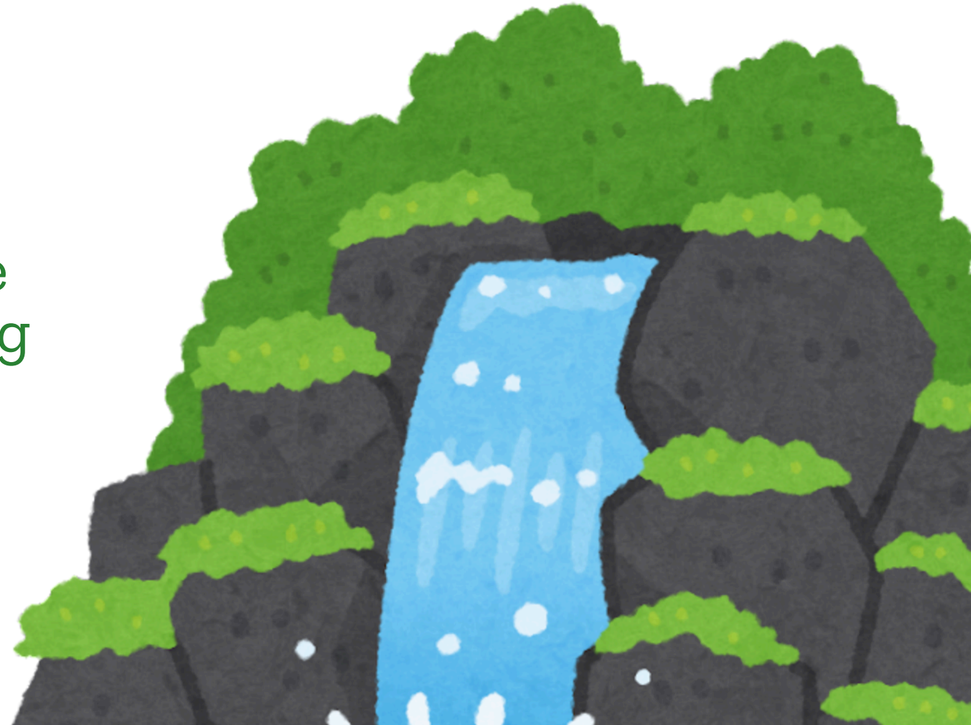
OTHER SERVICES → NOTIFICATION : ASYNCHRONOUS (RABBITMQ)

CONSUME RELEVANT BUSINESS EVENTS TO TRIGGER NOTIFICATIONS

CONCLUSION



- ReLoop is more than a recycling marketplace.
- It is a complete ecosystem that connects users, collectors, and recycling companies .
- Our project aims to make recycling easier, more accessible, and more rewarding while measuring its real environmental benefits.



THANK YOU !



